

Exercise 8

Consider the full conformal prediction with $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ being exchangeable and a symmetric score function s , prove that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} \leq 1 - \alpha + \mathbb{E} \left[\frac{1}{n+1} \sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}} \right]$$

where

$$\hat{q} = \text{Quantile}(S_1, \dots, S_{n+1}; 1 - \alpha).$$

Give an example showing that this bound is tighter than

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} \leq \frac{\lceil (1 - \alpha)(n + 1) \rceil}{n + 1} + \epsilon_{\text{tie}}$$

where ϵ_{tie} is the probability of S_{n+1} being equal to some S_j for $j \in \{1, \dots, n\}$.

Solution As already discussed in the lecture and in the textbook, $Y_{n+1} \in \mathcal{C}(X_{n+1})$ if and only if $S_{n+1} \leq \hat{q}$ and so

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} = P\{S_{n+1} \leq \hat{q}\} = P\{S_{n+1} < \hat{q}\} + P\{S_{n+1} = \hat{q}\}.$$

Given that $(S_1, \dots, S_n, S_{n+1})$ is exchangeable, $P\{S_{n+1} = \hat{q}\} = P\{S_i = \hat{q}\}$ for all $i \in \{1, \dots, n\}$ and

$$P\{S_{n+1} = \hat{q}\} = \frac{1}{n+1} \sum_{i=1}^{n+1} P\{S_i = \hat{q}\} = \mathbb{E} \left[\frac{1}{n+1} \sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}} \right].$$

The only thing left is to show that $P\{S_{n+1} < \hat{q}\} \leq 1 - \alpha$. To do so, first observe that $S_{n+1} < \hat{q}$ if and only if $S_{n+1} < S_{(k)}$, where $k = \lceil (1 - \alpha)(n + 1) \rceil$. Thus:

$$P\{S_{n+1} < \hat{q}\} = P\{S_{n+1} < S_{(k)}\} \leq \frac{\lceil (1 - \alpha)(n + 1) \rceil - 1}{n + 1} = \frac{\lfloor (1 - \alpha)(n + 1) \rfloor}{n + 1} \leq \frac{(1 - \alpha)(n + 1)}{n + 1} = 1 - \alpha.$$

This completes the proof.

Finally, in case when the random variables S_i 's are all continuous random variables, $\epsilon_{\text{tie}} = P\{S_{n+1} = \hat{q}\} = 0$ and evidently the bound that is worked out in this exercise is better. In case when the random variables S_i 's are discrete and can only take very limited values, it may happen that $S_1, \dots, S_n$ has taken all possible values and $\epsilon_{\text{tie}} = 1$ so that the bound worked out here is really the only available bound. ///